

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Vth Semester of BSc Biotechnology/Biochemistry/Microbiology Examination, March-2018

BS502: Concepts of Genetic Engineering

Date: 27/03/2018 Day: Tuesday Time: 01.30 pm To 01.50 pm

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - IA Choose the correct answer for the following questions.

5

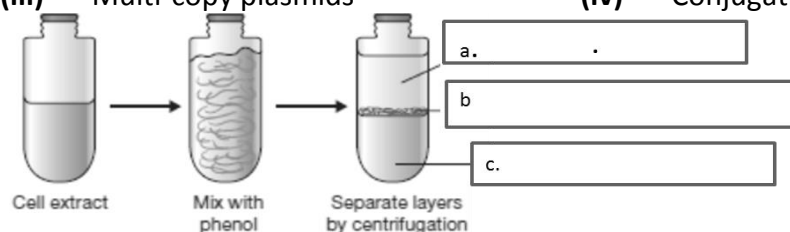
1. Typical PCR reaction requires

- | | |
|-------------------------|---------------------|
| (i) Klenow fragment | (ii) Taq Polymerase |
| (iii) DNA polymerase II | (iv) None of above |

2. The “R plasmid” which is exist in single copy in a cell can be known as

- | | |
|------------------------------|-------------------------------|
| (i) Low copy number plasmids | (ii) Incompatibility plasmids |
| (iii) Multi-copy plasmids | (iv) Conjugative plasmids |

3.



What “a”, “b” and “c” are representing in above figure?

- | | | | |
|------------------------|-------------------------|--------------------------|--------------------|
| (i) DNA-Protein-Phenol | (ii) Phenol-DNA-Protein | (iii) Protein-Phenol-DNA | (iv) None of above |
|------------------------|-------------------------|--------------------------|--------------------|

4. The given DNA sequence (5' ATGC 3') is incubated with enzyme Terminal

deoxynucleotide transferase and Adenine nucleotides, on which particular end the enzyme will add the adenine nucleotides?

- (i) 3' (ii) 5' (iii) Both 3' and 5' (iv) It is not a function of enzyme

5. From the following which is/are the type of lambda phage vectors?

- (i) Insertion (ii) Replacement (iii) Cosmid (iv) All of above

Q-IB Match the items of Column A with appropriate item of Column B

5

	A	B	Answer
6.	Ti Plasmid	A Droshophilla	
7.	P-elements	B Saccharomyces cerevisiae	
8.	YAC	C Transfer of DNA in to plant genome	
9.	DNASE I	D Cut the DNA away from Recognition sequence	
10.	Restriction endonuclease type I	E Nick single stranded DNA	

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Vth Semester of BSc Biotechnology/Biochemistry/Microbiology Examination, March-2018

BS502: Concepts of Genetic Engineering

Date: 27/03/2018 Day: Tuesday Time: 01.50 pm To 03.30 pm

Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed [Any two]

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- A** Describe the features of phage lambda suggesting its suitability as vector? What kind of plaques it gives? What is stuffer fragment? Briefly discuss Lambda replacement vectors
- B** Briefly discuss the method of transformation, transfection and electroporation and its role in gene transfer
- C** Discuss the concept of DNA libraries? Briefly describe the method for obtaining DNA libraries

SECTION - II

Q-III Answer the following questions as directed [Any Five]

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- A** Discuss the usefulness of genetic engineering, elaborate its application in development of better plant varieties.
- B** What is plasmid? Enlist the methods that used for its extraction from prokaryotic cell. Briefly describe any one from them.
- C** Write a note on Homopolymer tailing
- D** Discuss the method for direct selection of clones
- E** Give the name for any three markers that can be used for the selection of clone, briefly discuss the mechanism for one.
- F** How the restriction enzymes differ from other nucleases? Discuss its application in Genetic engineering
- G** Describe the role of linkers and adaptors in cloning